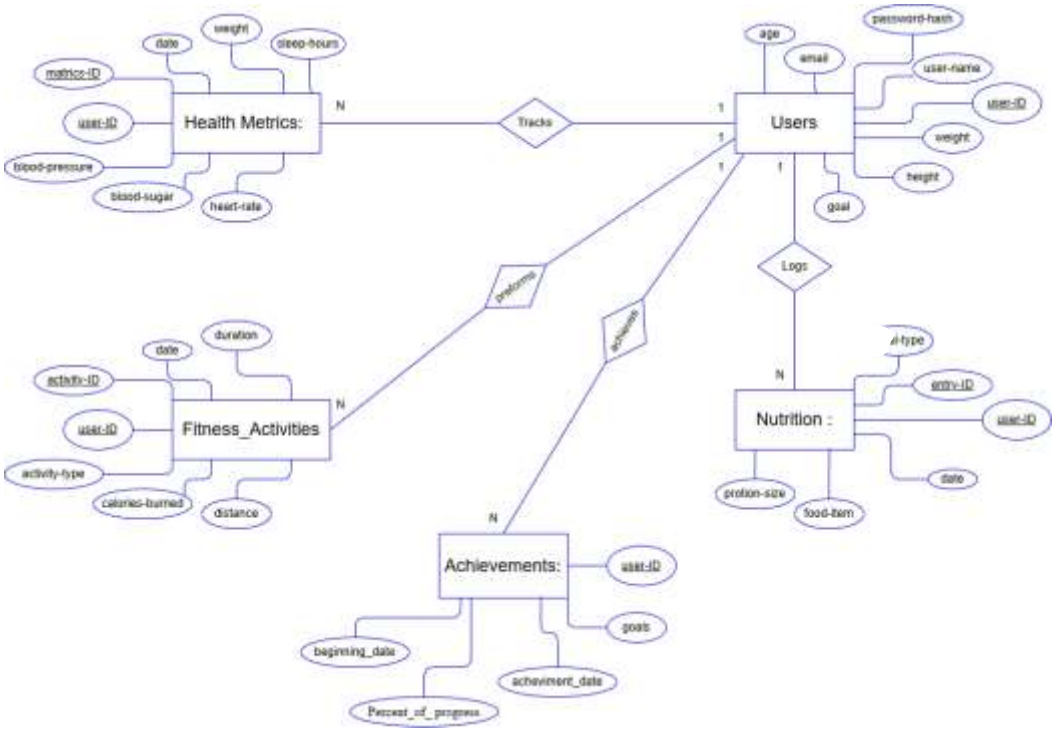
Health Tracker Applictaion

CEDS124

Problem and Overview: The need for people to regularly monitor their health has been highlighted by the growth in lifestyle diseases and chronic health issues. Nonetheless, a lot of people have trouble keeping track of their health indicators, comprehending their development, and making wise choices about their wellbeing. This gap offers an opportunity for a health tracking application that offers individualized insights and suggestions in addition to monitoring many health metrics.

The Health Tracker Application is a user-friendly app that helps people keep track of their health, fitness, and diet. It allows users to log into daily health information, record workouts, track meals, and see their progress and their achievements over time.

ER model:



This database has five entities: Users, Health Metrics, Fitness Activities, Achievements and Diet Entries

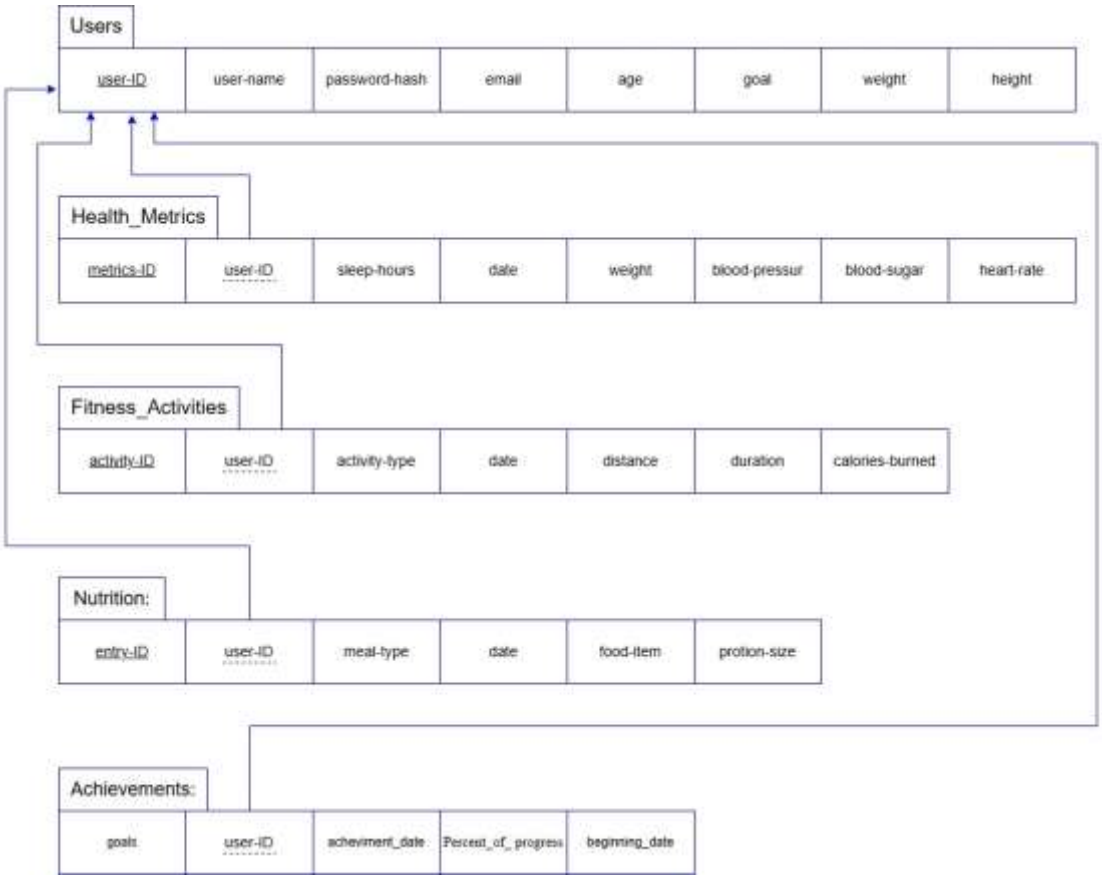
- **Users**: This entity has information about each user, including personal details like age, email, username, password, weight, height . Each user is uniquely identified by a user-ID.
- **Health Metrics**: This entity tracks the health of users, such as weight, blood pressure, blood sugar, heart rate, and sleep hours. Every record has a user-ID and is uniquely identified by matrices-ID.
- **Fitness Activities**: This entity logs multiple physical activities done by users, activity type, duration, and burning calories. Each fitness activity is performed by a user using user-ID and is uniquely identified by activity-ID.
- **Nutrition:** This table has information about user's diet habits, including meal type, portion size, food items, Each log in by the user is with a user-ID and is identified by entry-ID.

- **Achievements:** This entity records the user achievements , including their goal, Beginning date, Acheivement_date , Percent_of_ progress. Each record to the user is uniquely identified by user-ID.

The relationships:

- Each user can track multiple health metrics (1:M).
- Each user can perform multiple fitness activities (1:M).
- Each user can log multiple meals (1:M).
- Each user achieves multiple achievements (1:M).

Relational schema:



This database outlines the structure of the data of five entities: Users, Health Metrics, Fitness Activities, Achievements and Diet Entries.

1. **Users**:
   - user-ID: Unique identifier for each user (primary key).
   - username.
   - password-hash.
   - email.

- age, weight, height.

2. **Health Metrics:**
   - metrics-ID: Unique identifier for health records (primary key).
   - user-ID: references to the user-id in Users table (foreign key).
   - sleep-hours, date, weight, blood-pressure, blood-sugar, heart rate.

3. **Fitness Activities:**
   - activity-ID: Unique identifier for each activity entry (primary key).
   - user-ID: references to the user-id in Users table (foreign key).
   - activity-type.
   - date, distance, duration, calories burned.

4. **Nutrition:**
   - entry-ID: Unique identifier for each dietary log (primary key).
   - user-ID: references to the user-id in Users table (foreign key).
   - meal-type, date, food-item, portion-size.

5. **Achievements:**

- User_ID: references to the user-id in Users table (foreign key).
- Goal: references to the goal in Users table (foreign key).
- Beginning date, Achievement_ date, Percent_of_ progress

Normalization:

❖ 1NF – First Normal Form

First normal form is not needed since all of the fields are atomic and none of them is multivalued or nested. Each column in the users table holds only one value no lists, arrays, or sets.

❖ 2NF –Second Normal Form

Already done since every non-primary-key attribute is fully dependent on the entire primary key. All of the tables have a single column primary key.

❖ 3NF –Third Normal Form

All of the non-key attributes do not depend on other non-key attributes which means the 3rd normal form is already done.

Functional dependencies:

- **User table:**

user_ID->username, password, email, age, weight, height

- **Health Metrics table:**

metrics_ID ->user_id, date, weight, blood_pressure, heart_rate, blood_sugar, sleep_hours

user_ID, date → weight, blood_pressure, heart_rate, blood_sugar, sleep_hours

- **Fitness Activities table:**

activity_ID-> user_id, date, activity_type, duration, distance, calories_burned

- **Nutrition table:**

entry_ID -> user_id, date, meal_type, food_item, portion_size, calories

- **Achievements table:**

user_ID ->beginning_date, goal , achievement_date, percent_of_progress

- **Users table :**

User ID	Username	Password-hash	Email	Age	goal	weight	height
1	Mays	1e9b1c9f4f9c	Mays@gmail.com	18	Muscle gain	51 kg	159 cm
2	Lama	b0d9a7e802a	Lama@gmail.com	18	Streng-th	52 kg	161 cm
3	Jawan	c7b9a7ec3d2a	Jawan@gmail.com	18	Flexib-ility	50 kg	160 cm
4	Ghaliah	a0e4f1c6f3d7a	Ghaliah@gmail.com	18	Muscle gain	52 kg	160 cm
5	Rawnaa	e9c3a8f7b6d	Rawnaa@gmail.com	18	Streng-th	51 kg	161 cm

- **Health Metrics table :**

Metrics ID	User ID	Sleep Hours	The Date	Weight	Blood Pressure	Blood Sugar	Heart Rate
muscle_mass_kg	1	9	02-11-24	51 kg	100 mmHg	120 mg/dL	78 bpm
daily_steps	2	7	23-01-24	52 kg	90 mmHg	90 mg/dL	90 bpm
body_fat_pct	3	11	11-01-25	50 kg	112 mmHg	100 mg/dL	88 bpm
protein_intake_g	4	10	27-10-24	52 kg	97 mmHg	98 mg/dL	75 bpm
distance_traveled_km	5	8	19-02-25	51 kg	88 mmHg	111 mg/dL	95 bpm

- **Fitness Activities table:**

Activity_ID	User ID	Date	Activity_type	Duration	Distance	Calories_burned
6	1	02-11-24	Running	30 min	4 km	300
8	2	23-01-24	Swimming	45 min	1 km	500
7	3	11-01-25	Yoga	20 min	NULL	200
9	4	27-10-24	Hiking	60 min	5 km	600
10	5	19-02-25	Cycling	40 min	10 km	400

- **Nutrition table:**

Entry ID	User ID	Meal Type	The Date	Food item	Portion Size
101	1	Breakfast	02-11-24	Oatmeal	1 cup
102	2	Snack	23-01-24	Granola	150 grams
103	3	Dinner	11-01-25	Salad	2 cups
104	4	Lunch	27-10-24	Grilled chicken	30 grams
105	5	Breakfast	19-02-25	Smoothie	1 glass

- **Achievements Table:**

Goals	User ID	Achievement Date	Percent of progress	Beginning Date
Muscle gain	1	02-03-25	60%	02-11-24
Strength	2	23-06-24	100%	23-01-24
Flexibility	3	11-07-25	75%	11-01-25
Muscle gain	4	27-03-25	50%	27-10-24
Strength	5	19-09-25	80%	19-02-25

Implementation of the relational DB:

1. **Create the normalized tables:**

```sql
CREATE TABLE Users (
    User_ID INT PRIMARY KEY,
    Username VARCHAR(50),
    Password_Hash VARCHAR(255),
    Email VARCHAR(100),
    Age INT,
    Goal VARCHAR(50),
    Weight VARCHAR(10),
    Height VARCHAR(10)
);
CREATE TABLE HealthMetrics (
    Metrics_ID VARCHAR(100) PRIMARY KEY,
    User_ID INT,
    Sleep_Hours INT,
    TheDate DATE,
    Weight VARCHAR(10),
    Blood_Pressure VARCHAR(10),
    Blood_Sugar VARCHAR(10),
    Heart_Rate VARCHAR(10),
    FOREIGN KEY (User_ID) REFERENCES Users(User_ID)
);
CREATE TABLE FitnessActivities (
    Activity_ID INT PRIMARY KEY,
    User_ID INT,
    TheDate DATE,
    Activity_Type VARCHAR(50),
    Duration VARCHAR(10),
    Distance VARCHAR(10),
    Calories_Burned INT,
    FOREIGN KEY (User_ID) REFERENCES Users(User_ID)
);
```

```sql
CREATE TABLE Nutrition (
    Entry_ID INT PRIMARY KEY,
    User_ID INT,
    Meal_Type VARCHAR(50),
    TheDate DATE,
    FoodItem VARCHAR(100),
    PortionSize VARCHAR(50),
    FOREIGN KEY (User_ID) REFERENCES Users(User_ID)
);
 CREATE TABLE Achievements (
    Goals VARCHAR(50),
    User_ID INT,
    Achievement_Date DATE,
    PercentOfProgress VARCHAR(10),
    Beginning_Date DATE,
    FOREIGN KEY (User_ID) REFERENCES Users(User_ID)
);
```

2. Inserting data into the tables:

```sql
INSERT INTO Users (User_ID, Username, Password_Hash, Email, Age, Goal, Weight, Height) VALUES
(1, 'Mays', '1e9b1c94f9c', 'Mays@gmail.com', 18, 'Muscle gain', '51 kg', '159 cm');
INSERT INTO Users (User_ID, Username, Password_Hash, Email, Age, Goal, Weight, Height) VALUES
(2, 'Lama', '1a9b1e78e802', 'Lama@gmail.com', 21, 'Strength', '52 kg', '161 cm');
INSERT INTO Users (User_ID, Username, Password_Hash, Email, Age, Goal, Weight, Height) VALUES
(3, 'Jawan', 'c7b9a7ec3d2a', 'Jawan@gmail.com', 20, 'Flexibility', '50 kg', '160 cm');
INSERT INTO Users (User_ID, Username, Password_Hash, Email, Age, Goal, Weight, Height) VALUES
(4, 'Ghaliah', 'ae04f16cb7d7', 'Ghaliah@gmail.com', 19, 'Muscle gain', '52 kg', '161 cm');
INSERT INTO Users (User_ID, Username, Password_Hash, Email, Age, Goal, Weight, Height) VALUES
(5, 'Rawnaa', '89c83f8f6d7', 'Rawnaa@gmail.com', 18, 'Strength', '51 kg', '161 cm');


INSERT INTO HealthMetrics (Metrics_ID, User_ID, Sleep_Hours,TheDate, Weight, Blood_Pressure, Blood_Sugar, Heart_Rate) VALUES
('Muslce_mass_kg', 1, 9 , '02-NOV-24', '51 kg', '100 mmHg', '120 mg/dL', '78 bpm');
INSERT INTO HealthMetrics (Metrics_ID, User_ID, Sleep_Hours,TheDate, Weight, Blood_Pressure, Blood_Sugar, Heart_Rate) VALUES
('Daily_steps', 2, 7 , '23-JAN-24', '52 kg', '90 mmHg', '90 mg/dL', '90 bpm');
INSERT INTO HealthMetrics (Metrics_ID, User_ID, Sleep_Hours,TheDate, Weight, Blood_Pressure, Blood_Sugar, Heart_Rate) VALUES
('Body_fat_pct' , 3, 11, '11-JAN-25', '50 k', '112 mmHg', '100 mg/dL', '88 bpm');
INSERT INTO HealthMetrics(Metrics_ID, User_ID, Sleep_Hours, TheDate,Weight, Blood_Pressure, Blood_Sugar, Heart_Rate) VALUES
('Protein_intake_g' , 4, 10, '27-OCT-24' ,'52 kg', '97 mmHg', '98 mg/dL', '75 bpm');
INSERT INTO HealthMetrics (Metrics_ID, User_ID, Sleep_Hours, TheDate,Weight, Blood_Pressure, Blood_Sugar, Heart_Rate) VALUES
('Distance_traveled_km' , 5, 8 , '19-FEB-25', '51 kg', '88 mmHg', '111 mg/dL', '95 bpm');
```

```sql
INSERT INTO FitnessActivities (Activity_ID, User_ID, TheDate, Activity_Type, Duration, Distance, Calories_Burned) VALUES
(6, 1, '02-NOV-24', 'Running', '30 min', '4 km', 300);
INSERT INTO FitnessActivities (Activity_ID, User_ID, TheDate, Activity_Type, Duration, Distance, Calories_Burned) VALUES
(8, 2, '23-JAN-24', 'Swimming', '45 min', '1 km', 500);
INSERT INTO FitnessActivities (Activity_ID, User_ID, TheDate, Activity_Type, Duration, Distance, Calories_Burned) VALUES
(7, 3, '11-JAN-25', 'Yoga', '20 min', NULL, 200);
INSERT INTO FitnessActivities (Activity_ID, User_ID, TheDate, Activity_Type, Duration, Distance, Calories_Burned) VALUES
(9, 4, '27-OCT-24', 'Hiking', '60 min', '5 km', 600);
INSERT INTO FitnessActivities (Activity_ID, User_ID, TheDate, Activity_Type, Duration, Distance, Calories_Burned) VALUES
(10, 5, '19-FEB-25', 'Cycling', '40 min', '10 km', 400);


INSERT INTO Nutrition (Entry_ID, User_ID, Meal_Type, TheDate, FoodItem, PortionSize) VALUES
(101, 1, 'Breakfast', '02-NOV-24', 'Oatmeal', '1 cup');
INSERT INTO Nutrition (Entry_ID, User_ID, Meal_Type, TheDate, FoodItem, PortionSize) VALUES
(102, 2, 'Snack', '23-JAN-24', 'Granola', '150 grams');
INSERT INTO Nutrition (Entry_ID, User_ID, Meal_Type, TheDate, FoodItem, PortionSize) VALUES
(103, 3, 'Dinner', '11-JAN-25', 'Salad', '2 cups');
INSERT INTO Nutrition (Entry_ID, User_ID, Meal_Type, TheDate, FoodItem, PortionSize) VALUES
(104, 4, 'Lunch', '27-OCT-24', 'Grilled chicken', '30 grams');
INSERT INTO Nutrition (Entry_ID, User_ID, Meal_Type, TheDate, FoodItem, PortionSize) VALUES
(105, 5, 'Breakfast', '19-FEB-25', 'Smoothie', '1 glass');


INSERT INTO Achievements (Goals, User_ID, Achievement_Date, PercentOfProgress, Beginning_Date) VALUES
('Muscle gain', 1, '02-MAR-25', '60', '02-NOV-24');
INSERT INTO Achievements (Goals, User_ID, Achievement_Date, PercentOfProgress, Beginning_Date) VALUES
('Strength', 2, '23-JUN-24', '100', '23-JAN-24');
INSERT INTO Achievements (Goals, User_ID, Achievement_Date, PercentOfProgress, Beginning_Date) VALUES
('Flexibility', 3, '11-JUL-25', '75', '11-JAN-25');
INSERT INTO Achievements (Goals, User_ID, Achievement_Date, PercentOfProgress, Beginning_Date) VALUES
('Muscle gain', 4, '27-MAR-25', '50', '27-OCT-24');
INSERT INTO Achievements (Goals, User_ID, Achievement_Date, PercentOfProgress, Beginning_Date) VALUES
('Strength', 5, '19-SEP-25', '80', '19-FEB-25');
```

```sql
SELECT*FROM Users;
```

USER_ID	USERNAME	PASSWORD_HASH	EMAIL	AGE	GOAL	WEIGHT	HEIGHT
1	Mays	1e9b1c94f9c	Mays@gmail.com	18	Muscle gain	51 kg	159 cm
2	Lama	1a9b1e78e802	Lama@gmail.com	21	Strength	52 kg	161 cm
3	Jawan	c7b9a7ec3d2a	Jawan@gmail.com	20	Flexibility	50 kg	160 cm
4	Ghaliah	ae04f16cb7d7	Ghaliah@gmail.com	19	Muscle gain	52 kg	161 cm
5	Rawnaa	e9c83f8f6d7	Rawnaa@gmail.com	18	Strength	51 kg	161 cm

```sql
SELECT *FROM HealthMetrics;
```

METRICS_ID	USER_ID	SLEEP_HOURS	THEDATE	WEIGHT	BLOOD_PRESSURE	BLOOD_SUGAR	HEART_RATE
Muslce_mass_kg	1	9	02-NOV-24	51 kg	100 mmHg	120 mg/dL	78 bpm
Daily_steps	2	7	23-JAN-24	52 kg	90 mmHg	90 mg/dL	90 bpm
Body_fat_pct	3	11	11-JAN-25	50 k	112 mmHg	100 mg/dL	88 bpm
Protein_intake_g	4	10	27-OCT-24	52 kg	97 mmHg	98 mg/dL	75 bpm
Distance_traveled_km	5	8	19-FEB-25	51 kg	88 mmHg	111 mg/dL	95 bpm

```sql
SELECT*FROM FitnessActivities;
```

ACTIVITY_ID	USER_ID	THEDATE	ACTIVITY_TYPE	DURATION	DISTANCE	CALORIES_BURNED
6	1	02-NOV-24	Running	30 min	4 km	300
8	2	23-JAN-24	Swimming	45 min	1 km	500
7	3	11-JAN-25	Yoga	20 min	-	200
9	4	27-OCT-24	Hiking	60 min	5 km	600
10	5	19-FEB-25	Cycling	40 min	10 km	400

```sql
SELECT*FROM Nutrition;
```

ENTRY_ID	USER_ID	MEAL_TYPE	THEDATE	FOODITEM	PORTIONSIZE
101	1	Breakfast	02-NOV-24	Oatmeal	1 cup
102	2	Snack	23-JAN-24	Granola	150 grams
103	3	Dinner	11-JAN-25	Salad	2 cups
104	4	Lunch	27-OCT-24	Grilled chicken	30 grams
105	5	Breakfast	19-FEB-25	Smoothie	1 glass

```sql
SELECT*FROM Achievements;
```

GOALS	USER_ID	ACHIEVEMENT_DATE	PERCENTOFPROGRESS	BEGINNING_DATE
Muscle gain	1	02-MAR-25	60	02-NOV-24
Strength	2	23-JUN-24	100	23-JAN-24
Flexibility	3	11-JUL-25	75	11-JAN-25
Muscle gain	4	27-MAR-25	50	27-OCT-24
Strength	5	19-SEP-25	80	19-FEB-25

3. Design and implement 4 queries :

1) Query to retrieve data from table:

```sql
SELECT
    HealthMetrics.Metrics_ID,
    HealthMetrics.weight,
    HealthMetrics.blood_pressure,
    HealthMetrics.blood_sugar,
    HealthMetrics.heart_rate,
    HealthMetrics.sleep_hours
FROM
    HealthMetrics
WHERE
  HealthMetrics.User_ID = 1;
```

METRICS_ID	WEIGHT	BLOOD_PRESSURE	BLOOD_SUGAR	HEART_RATE	SLEEP_HOURS
Muslce_mass_kg	51 kg	100 mmHg	120 mg/dL	78 bpm	9

2) Query to sum calories burned by users with group by:

```sql
SELECT
    Users.Username,
    SUM(FitnessActivities.Calories_Burned) AS total_Calories_Burned
FROM
    Users
JOIN
    FitnessActivities  ON Users.User_ID = FitnessActivities.User_ID
GROUP BY
    Users.Username;
```

USERNAME	TOTAL_CALORIES_BURNED
Jawan	200
Rawnaa	400
Mays	300
Lama	500
Ghaliah	600

3) Query that find users with achievements above 75% with ORDER BY:

```sql
SELECT
    Users.Username ,
    Achievements.Goals,
    Achievements.PercentOfProgress
FROM
    Users
JOIN
    Achievements ON Users.User_ID = Achievements.User_ID
WHERE
    Achievements.PercentOfProgress > 75
ORDER BY
    Achievements.PercentOfProgress DESC;
```

USERNAME	GOALS	PERCENTOFPROGRESS
Rawnaa	Strength	80
Lama	Strength	100

4)Query to find the average number of calories burned by all users using SUBQUERY:

```sql
SELECT AVG(Calories_Burned) AS Avg_Calories_Burned
FROM FitnessActivities
WHERE User_ID IN (
  SELECT User_ID
  FROM Users
);
```

AVG_CALORIES_BURNED
400

Design stored procedures:

1)

```
CREATE OR REPLACE PROCEDURE UpdateFitnessActivity (
      InputActivityID IN NUMBER,
    InputTheDate IN DATE,
    InputActivityType IN VARCHAR2,
    InputDuration IN VARCHAR2,
    InputDistance IN VARCHAR2,
    InputCaloriesBurned IN NUMBER
)
IS
BEGIN
    UPDATE FitnessActivities
    SET
        TheDate = InputTheDate,
        Activity_Type = InputActivityType,
        Duration = InputDuration,
        Distance = InputDistance,
        Calories_Burned = InputCaloriesBurned
    WHERE
        Activity_ID = InputActivityID;
END;
/
```

```
Procedure created.
```

SQL Worksheet

```
exec UpdateFitnessActivity( 6, TO_DATE('11-JAN-25', 'DD-MON-YY'),'Running','45 min','5 km',600);
```

```
Statement processed.
```

```
SELECT *
FROM FitnessActivities
WHERE user_id = 1;
```

ACTIVITY_ID	USER_ID	THEDATE	ACTIVITY_TYPE	DURATION	DISTANCE	CALORIES_BURNED
6	1	11-JAN-25	Running	45 min	5 km	600

(2

```oracle
CREATE OR REPLACE PROCEDURE GetHealthMetricsByUserID (InputUserID IN NUMBER)
AS
BEGIN
    FOR Hmetrics IN (
        SELECT Metrics_ID, Weight, Blood_Pressure, Blood_Sugar, Heart_Rate, Sleep_Hours
        FROM HealthMetrics
        WHERE User_ID = InputUserID
    )
    LOOP
        DBMS_OUTPUT.PUT_LINE('Metrics ID: ' || Hmetrics.Metrics_ID);
        DBMS_OUTPUT.PUT_LINE('Weight: ' || Hmetrics.Weight);
        DBMS_OUTPUT.PUT_LINE('Blood Pressure: ' || Hmetrics.Blood_Pressure);
        DBMS_OUTPUT.PUT_LINE('Blood Sugar: ' || Hmetrics.Blood_Sugar);
        DBMS_OUTPUT.PUT_LINE('Heart Rate: ' || Hmetrics.Heart_Rate);
        DBMS_OUTPUT.PUT_LINE('Sleep Hours: ' || Hmetrics.Sleep_Hours);
    END LOOP;
END GetHealthMetricsByUserID;
```

```
Procedure created.
```

```oracle
exec GetHealthMetricsByUserId (1);
```

```
Statement processed.
Metrics ID: Muslce_mass_kg
Weight: 51 kg
Blood Pressure: 100 mmHg
Blood Sugar: 120 mg/dL
Heart Rate: 78 bpm
Sleep Hours: 9
```